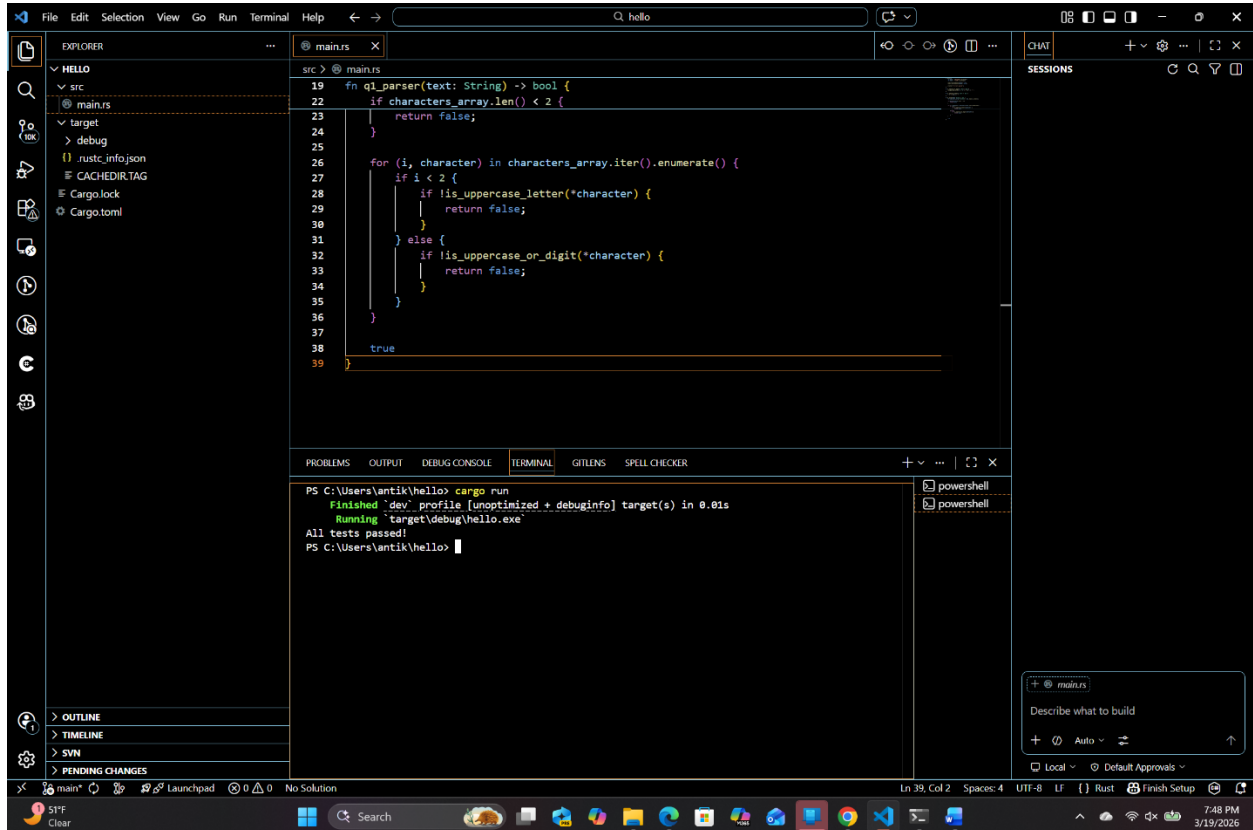


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Course: COSC 455

Assignment: Lab 3



The screenshot shows an IDE with a dark theme. The Explorer panel on the left shows a project structure with folders 'src' and 'target', and files 'Cargo.lock' and 'Cargo.toml'. The main editor displays a Rust function `fn q1_parser(text: String) -> bool` that checks if a string contains at least one uppercase letter or digit. The terminal at the bottom shows the command `cargo run` being executed, with output indicating that the tests passed. The status bar at the bottom shows 'Ln 39, Col 2, Spaces: 4, UTF-8, LF, Rust, Finish Setup'.

```
src > @ main.rs
19 fn q1_parser(text: String) -> bool {
22     if characters_array.len() < 2 {
23         return false;
24     }
25
26     for (i, character) in characters_array.iter().enumerate() {
27         if i < 2 {
28             if !is_uppercase_letter(*character) {
29                 return false;
30             }
31         } else {
32             if !is_uppercase_or_digit(*character) {
33                 return false;
34             }
35         }
36     }
37
38     true
39 }
```

```
PS C:\Users\antik\hello> cargo run
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.01s
Running `target\debug\hello.exe`
All tests passed!
PS C:\Users\antik\hello>
```

1. Parsing the input “teh kat lovez a dawg”

Parsing begins by calling `lolspeak().lolspeak()` method first calls `noun_phrase()`. Inside `noun_phrase()`, the parser calls `article()`. The current token is “teh”, so `article()` recognizes and consumes it. Next, `noun_phrase()` calls `noun()`. The current token is “kat”, so `noun()` recognizes and consumes it. The first noun phrase is now complete. Control returns to `lolspeak()`, which then calls `verb()`. The current token is “lovez”, and since it is a valid verb, `verb()` recognizes and consumes it.

Next, `lolspeak()` calls `noun_phrase()` again. Inside this method, `article()` is called and consumes “a”, since it is a valid article. Then `noun()` is called and consumes “dawg”, since it is a valid noun. At this point, all tokens have been consumed in the correct order, and parsing succeeds.

Order of method calls: `lolspeak` → `noun_phrase` → `article` → `noun` → `verb` → `noun_phrase` → `article` → `noun`

2. Explanation of Methods

`lolspeak()` This method succeeds only if the input follows the structure of a noun phrase, followed by a verb, followed by another noun phrase. This method reports a syntax error when the input does not match this order or when any required component is missing or invalid.

This method corresponds to the grammar rule: `<lolspeak> ::= <noun_phrase> <verb> <noun_phrase>`

`noun_phrase()`

This method succeeds only if the input starts with a valid article followed by a valid noun. This method reports a syntax error when the first token is not an article or when the following token is not a valid noun. This method corresponds to the grammar rule: `<noun_phrase> ::= <article> <noun> verb()`

This method succeeds only if the current token is a valid verb such as `lovez`, `hatez`, or `ates`. This method reports a syntax error when the current token is not one of the valid verbs. This method corresponds to the grammar rule:

`<verb> ::= lovez | hatez | ates`

3. Parsing Results

Input 1: a rat lovez teh kat

Parsing succeeds.

The first noun phrase is “a rat”, which is valid. The verb “lovez” is valid. The second noun phrase “teh kat” is also valid. The input follows the correct grammar structure, so parsing succeeds.

Input 2: teh dawg a lovez rat

Parsing fails.

The parser starts correctly by recognizing “teh dawg” as a noun phrase. After this, the parser expects a verb. However, the next token is “a”, which is an article, not a verb.

The error occurs in the verb() method because it expected a valid verb but found “a” instead. This violates the grammar rule, so parsing fails.

Input 3: teh kat runs a rat

Parsing fails.

The parser correctly recognizes “teh kat” as a noun phrase. It then expects a verb. The next token is “runs”, which is not one of the valid verbs in the grammar. The error occurs in the verb() method because it expected lovez, hatez, or ates, but found “runs” instead. Therefore, parsing fails.